BEING SYNTOMIC IS SYNTOMIC LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof line 7125. Reduces the claim to four clauses.
- Zariski locality on source and target and preservation under precomposition for syntomic morphisms. Morphisms, Lemmas morphisms-lemma-syntomic-characterize and morphisms-lemma-composition-syntomic, invoked at proof lines 7126-7128. Supply clauses (1)-(3).
- Permanence of syntomicity under a syntomic surjection on the source. Same chapter, Lemma lemma-syntomic-smooth-etale-permanence, invoked at proof line 7129. Supplies clause (4).

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KM: sources/stacks-descent/pinned-original-source.tex, source lines 4640–4640. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UH: sources/stacks-chow/tags.txt, source lines 2410–2410. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UD: sources/stacks-chow/tags.txt, source lines 2406–2406. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *The property $\mathcal{P}(f) =$"f is syntomic" is syntomic local on the source.*

Proof. Combine Lemma 036H with Morphisms, Lemma 01UD (local for Zariski on source and target), Morphisms, Lemma 01UH (pre-composing), and Lemma 02KM (part (4)). □